

Rep Smarter, Not Harder: AI Hypertrophy Coaching with Wearable Sensors and Edge Neural Networks

1st Musa Azeem

College of Engineering and Computing
University of South Carolina
Columbia, SC, USA
mmazeem@email.sc.edu

2nd Grant King

College of Engineering and Computing
University of South Carolina
Columbia, SC, USA
gtking@email.sc.edu

3rd Savannah Noblitt

College of Engineering and Computing
University of South Carolina
Columbia, SC, USA
snoblitt@email.sc.edu

Abstract—abstract

Index Terms—component, formatting, style, styling, insert

I. INTRODUCTION

A. Background

Exercise science background

B. Related Work

Literature review

C. Contributions

What we did new

II. MATERIALS AND METHODS

A. Data Collection

1) *Participants*: Participants consisted of this work's authors and volunteers at three different gyms. There were 13 participants in total (9 male, 4 female). All participants were about 18-25 years old and varied in weight, height, strength, and resistance training experience.

2) *Equipment*: Participant preacher curls were recorded using a custom-made recording device, consisting of a Raspberry Pi 5 and the Adafruit MPU-6050 6-axis inertial measurement unit (IMU), capturing both accelerometer and gyroscope data. Although only IMU data was used here, the device, shown in Figure 1 was designed with the intention and capability of recording data from multiple sensors (such as heartbeat sensor and sEMG). The Raspberry Pi was connected to and accessed by a laptop. Three different preacher curl machines at the three locations were used for data collection: (TODO).

3) *Procedure*: During data collection, the IMU was taped to each participant's left wrist. The axes of the sensor were oriented as shown in Figure 2. Each participant was instructed to perform sets of preacher curls with consistent form, taking each set to momentary muscular failure. Participants were free to select the weight they lifted and decide how many sets they performed. Consequently, collected data varied in the amount of weight lifted, the number of repetitions per set, and how many sets were completed in a session. Participants also introduced natural variation in form and range of motion (e.g. locking out their arms at the bottom of the curl, duration of pause at the bottom, height of the concentric phase). Only

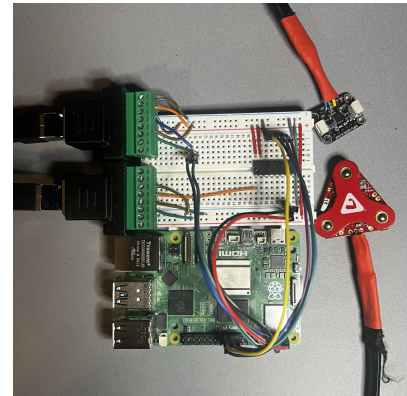


Fig. 1. **Data collection device.** The Raspberry Pi (bottom) reads, processes, and stores data from the IMU (top right). The sEMG sensor (bottom right) was not used due to hardware issues. During data collection, the Pi was powered by a portable battery pack. A box was used to protect the Raspberry Pi from damage, and the IMU was taped to participants' wrists.

sets taken to failure with proper form were admitted to this work's dataset.

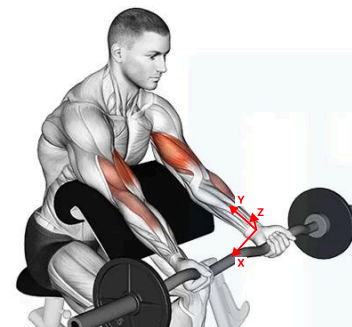


Fig. 2. **IMU orientation.** The IMU was taped to the participant's wrist with the axes oriented as shown. Image source [1].

During data collection, annotations were recorded for future labelling. This was facilitated by a custom Python server running on the Raspberry Pi. The server broadcasts an interactive data collection web application for access via laptop over wifi or ethernet. This data collection dashboard was used to enter participant metadata and location metadata, start the recording of

IMU data, manually record timestamp markers to annotate the end of each repetition, and stop the recording. The timestamp markers were later used to generate data labels.

B. Data Processing

data cleaning, windowing, splitting, augmentation

C. Segmentation Model Design

D. Classification Model Design

E. Model Training

[2]

F. Model Evaluation

G. Edge Deployment

describe steps taken to deploy to edge

III. RESULTS

Put any figures and tables here

A. Segmentation Model Performance

B. Classification Model Performance

C. Edge Deployment Performance

IV. DISCUSSION

REFERENCES

REFERENCES

- [1] K. Kings, "Best preacher curl alternatives," accessed April 30, 2025. [Online]. Available: <https://www.kettlebellkings.com/blogs/default-blog/best-preacher-curl-alternatives>
- [2] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2016, pp. 770–778.